ST2334 Cheat Sheet, AY26/27 S1

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November 10, 2026

1 Basic Concepts of Probability

1.1 Probability

Probability of an event A is a measure of the likelihood of the event occurring.

Relative frequency of A, when experiment is repeated n times and event A occurs m times, approaches $P(A)$ as $n \to \infty$:

$$P(A) = \lim_{\{n\to\infty\}} \frac{m}{n}$$

$$0 \le P(A) \le 1$$

$$P(S) = 1$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

1.2 Conditional Probability

Conditional Probability ($P(A \mid B)$) is probability of A occurring given that B occurred.

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B|A)}{P(B)}$$

Simpson's Paradox

Phenomenon where a trend appears in different groups of data but disappears or reverses when the groups are combined. Or formally,

$$P(A \mid B \cap C_i) \ge P(A \mid B' \cap C_i) \forall i \text{ but } P(A \mid B) < P(A \mid B')$$

Multiplication Rule

$$P(A \cap B) = P(A \mid B)P(B) \text{ if } P(B) > 0$$

1.3 Independence

Events A and B are **independent**, $A \perp B \Leftrightarrow$

$$P(A \cap B) = P(A)P(B)$$

Tautologies:

- Suppose P(A) > 0 and P(B) > 0, If $A \perp B$, then A and B are not mutually exclusive (aka $A \cap B \neq \varnothing$)
- $S \perp A$ for all events A; $\varnothing \perp A$ for all events A
- If $A \perp B$, then $A \perp B'$, $A' \perp B$, $A' \perp B'$
- Independence cannot be expressed in terms of venn diagram

1.4 Bayes' Theorem

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B \mid A)P(A)}{\sum_{\{i=1\}}^{n} P(B \mid A_i)P(A_i)}$$

Law of Total Probability

Suppose $A_1, A_2, ..., A_n$ is a partition of the sample space S

$$P(B) = \sum_{\{i=1\}}^{n} P(B \cap A_i) = \sum_{\{i=1\}}^{n} P(B \mid A_i)P(A_i)$$

$$P(B) = P(A)P(B \mid A) + P(A')P(B \mid A')$$

2 Random Variable

Random Variable is a function that transform a sample from sample space to a real number.

$$X : S \to \mathbb{R} : s \in S \text{ and } X(s) = x \in \mathbb{R}$$

Range space of X is set of real numbers that satisfies

$$R_X = \{x \mid X(s) = x \forall s \in S\} : R_X \in \mathbb{R}$$

Subsets of Sample Space

$$\{X = x\} = \{s \in S : X(s) = x\}, \{X = x\} \in S$$
$$\{X \in A\} = \{s \in S : X(s) \in A\}, \{X \in A\} \in S$$

2.1 Probability Distribution

Probability Mass Function (pmf)

Discrete random variable X_D, when $R_X = \{x_1, x_2, ...\}$ is finite or countable.

pmf, $f(x)$ for $\{X = x\}$

$$f(x) = \begin{cases} P(X = x) \mid \forall x \in R_X \\ 0 \mid \forall x \notin R_X \end{cases}$$

Well-defined pmf for discrete random variable X:

1. $\forall x_i \in R_X, f(x_i) \ge 0$
2. $\forall x_i \notin R_X, f(x_i) = 0$
3. $\sum_{i=1}^{\infty} f(x_i) = \sum_{x_i \in R_X} f(x_i) = 1$

Probability Density Function (pdf)

Continuous random variable X_C, when $R_X \in [a, b]$ is an interval(s).

pdf, $P(a \le X \le b)$ for any a and b such that $a \le b$

$$P(a \le X \le b) = \int_a^b f(x)dx$$

Well-defined pdf for continuous random variable X:

1. $\forall x \in R_X, f(x) \ge 0$
2. $\forall x \notin R_X, f(x) = 0$
3. $\int_{R_X} f(x)dx = 1$

Other key properties,

- Specific point is zero: $P(X = x_i) = \int_{x_i}^{x_i} f(x)dx = 0$
- Endpoint doesn't matter: $P(a \le X \le b) = P(a < X < b)$

2.2 Cumulative Distribution Function

Cumulative Distribution Function (cdf), F(x) for any random variable X is defined as

$$F(x) = P(X \le x) = \begin{cases} f(0) + f(1) + ... + f(x) \text{ if X is discrete} \\ \int_{-\infty}^{x} f(t)dt \text{ if X is continuous} \end{cases}$$

$$P(a \le X \le b) = P(X \le b) - P(X < a) = F(b) - F(a-)$$
$$P(X = x) = P(x \le X \le x) = F(x) - F(x-)$$

Let $F(x) = \int_{-\infty}^{x} f(t)dt$

$$f(x) = F'(x) = \frac{d}{dx}\int_{-\infty}^{x} f(t)dt$$

2.3 Expectation and Variance

Expectations or mean of X is defined by: (μ_X may not be in R_X)

$$\mu_X = E(X) = \sum_{x_i \in R_X} x_i f(x_i) = \int x f(x)dx$$

Properties of Expectation:

1. $E(aX + b) = aE(X) + b$
2. $E(X + Y) = E(X) + E(Y)$
3. $E[g(X)] = \sum_{x \in R_X} g(x)f(x)$ or $\int g(x)f(x)dx$
4. $E(x^2) = \sum x^2 f(x)$

Variance

Variance of X is defined as σ_X^2

$$\sigma_X^2 = V(X) = E\left[(X - \mu_X)^2\right] = E(X^2) - [E(X)]^2$$
$$= \sum_{x \in R_X} (x - \mu_X)^2 f(x) \text{ or } \int_{-\infty}^{\infty} (x - \mu_X)^2 f(x)dx$$

Standard deviation of X is defined as σ_X

$$\sigma_X = \sqrt{V(X)}$$

Properties of Variance:

- $V(aX + b) = a^2 V(X)$

3 Joint Distribution

- **n-dimentional random variable, $(X, Y, ...)$** $= (X(s), Y(s), ...) \forall s \in S$
- **Range Space** $R_{X,Y} = \{(x, y) \mid x = X(s), y = Y(s), \forall s \in S\}$
- **Discrete / Continuous** two-dimensional random variable depends if **both** random variable is countable and finite or vice-versa.

3.1 Joint, Marginal and Conditional Distribution

Joint probability (mass) function, jpf is defined by $(x, y) \in R_{X,Y}$

$$f_{X,Y}(x, y) = P(X = x, Y = y)$$

Well-defined jpf X, Y of both discrete and continuous random variable satisfies:

1. $f_{X,Y}(x, y) \ge 0, \forall (x, y) \in R_{X,Y}$
2. $f_{X,Y}(x, y) = 0, \forall (x, y) \notin R_{X,Y}$
3. $\sum_{i=1}^{\infty}\sum_{j=1}^{\infty} f_{X,Y}(x_i, y_i) = \sum_{i=1}^{\infty}\sum_{j=1}^{\infty} P(X = x_i, Y = y_j) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{X,Y}(x, y)dxdy = 1$

Marginal Probability Distribution

Marginal Probability Distribution of X is defined as sum of $f(y)$ after fixing $X = x$

$$f_X(x) = \sum_y f_{X,Y}(x, y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)dy$$

- $f_X(x)$ satisfies all properties of probability function

Conditional probability Function

Conditional Probability function of Y given $X = x$ is defined as

$$f_{Y \mid X}(y \mid x) = \frac{f_{X,Y}(x, y)}{f_X(x)}$$

- $f_{Y \mid X}(y \mid x)$ is defined only for x such that $f_X(x) > 0$
- $f_{Y \mid X}(y \mid x)$ is not a probability function and therefore, sum $\neq 1$

Applications:

- $P(Y \le y \mid X = x) = \int_{-\infty}^{y} f_{Y|X}(t|x)dt$
- Regression function $E(Y \mid X = x) = \int_{-\infty}^{\infty} y f_{Y|X}(y|x)dy$

3.2 Independent Random Variables

Random variable $X \perp Y$ are **independent** $\Leftrightarrow$

$$f_{X,Y}(x, y) = f_X(x)f_Y(y)$$
$\Leftrightarrow$ both of the following holds

1. $R_{X,Y}$ spans **Product Space** $R_{X,Y} = \{(x, y) \mid x \in R_X, y \in R_Y\} = R_X \times R_Y$, when probability function is positive.
2. $\forall (x, y) \in R_{X,Y}$

$$f_{X,Y}(x, y) = c \cdot g_1(x) \cdot g_2(y)$$

where g_1 depends only on x, g_2 depends only on y, and c is constant

Let $X \perp Y$

- $P(X \in A; Y \in B) = P(X \in A)P(Y \in B)$
- $P(X \le x; Y \le y) = P(X \le x)P(Y \le y)$
- Let $g1(.), g2(.)$ be arbitrary functions, $g1(X), g2(Y)$ are independent
- $f_X(x) > 0 \Rightarrow f_{Y \mid X}(y \mid x) = f_Y(y)$

3.3 Expectation, Covariance and Correlation Coefficient

Expectation, $E(g(X, Y))$:

$$E(g(X, Y)) = \sum_x \sum_y g(x, y)f_{X,Y}(x, y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x, y)f_{X,Y}(x, y)dydx$$

Covariance, $\text{cov}(X, Y)$

$$\text{cov}(X, Y) = \sum_x \sum_y (x - \mu_x)(y - \mu_y)f_{X,Y}(x, y)$$
$$= E[(X - \mu_X)(Y - \mu_Y)] = E(XY) - \mu_X\mu_Y$$

Properties of covariance:

- $X \perp Y \Rightarrow \text{cov}(X, Y) = 0$, converse is not true
- $X \perp Y \Rightarrow E(XY) = E(X)E(Y)$
- $\text{cov}(aX + b, cY + d) = ac \cdot \text{cov}(X, Y)$
- $\text{cov}(X, Y) = \text{cov}(Y, X)$

Correlation Coefficient, $\rho(X, Y)$

$$\rho(X, Y) = \frac{\text{cov}(X, Y)}{\sigma_X\sigma_Y}, -1 \le \rho(X, Y) \le 1$$

Properties of joint-variance:

- $V(X \pm Y) = V(X) + V(Y) \pm 2\text{ cov}(X, Y)$
- $V(aX + bY) = a^2V(X)b^2V(Y) - 2ab\text{ cov}(X, Y)$
- $V(X_1 + X_2 + ... + X_n) = V(X_1) + V(X_2) + ... + V(X_n) + 2\sum_{j>i} \text{cov}(X_i, X_j)$

4 Probability Distribution

4.1 Discrete Probability Distribution

$q = 1 - p$

Distribution	pmf	cdf	$E(X), V(X)$
Discrete uniform $U(k)$	$f(x) = \frac{1}{k}, x \in R_x$	$F(x) = \frac{\lvert\{x_i \le x\}\rvert}{k}$	$E(X) = \frac{\sum_i x_i}{k}$ $V(X) = \frac{\sum_i x_i^2}{k} - \mu^2$
Bernoulli $X \in \{0, 1\}$	$f(x) = p^x q^{1-x}$	$F(0) = q, F(1) = 1$	$E(X) = p$ $V(X) = pq$
Bin(n, p) # successes	$f(x) = \binom{n}{x}$ $p^x q^{n-x}$	$F(x) = \sum_{i=0}^{x}\binom{n}{i}$ $p^i q^{n-i}$	$E(X) = np$ $V(X) = npq$
NB(k, p) # trials to k successes	$f(x) = \binom{x-1}{k-1}$ $p^k q^{x-k}$	$F(x) = \sum_{i=k}^{x}\binom{i-1}{k-1}$ $p^k q^{i-k}$	$E(X) = \frac{k}{p}$ $V(X) = \frac{kq}{p^2}$
Geom(p) = NB$(1, p)$ # trials to 1st success	$f(x) = q^{x-1}p$	$F(x) = 1 - q^x$	$E(X) = \frac{1}{p}$ $V(X) = \frac{q}{p^2}$
Poisson(λ) # events in an interval	$f(x) = \frac{e^{-\lambda}\lambda^x}{x!}$	$F(x) = \sum_{i=0}^{x}\frac{e^{-\lambda}\lambda^i}{i!}$	$E(X) = \lambda$ $V(X) = \lambda$
Poisson process $\lambda = \alpha T$	same as Poisson	same as Poisson	$E(X) = \alpha T$ $V(X) = \alpha T$

Approximate Binomial using Poisson Distribution

As $n \to \infty$ and $p \to 0$ but $\lambda = np$ remains constant, We may approximate $X \sim \text{Bin}(n, p)$ with $X \sim \text{Poisson}(n, p)$

$$\lim_{p\to 0; n\to\infty} P(X = x) = \frac{e^{-np}(np)^x}{x!}$$

Condition to use:

$$n \ge 20 \wedge p \le 0.05 \text{ or } n \ge 100 \wedge np \le 10$$

4.2 Continuous Probability Distribution

Continuous Uniform Distribution, $X \sim U(a,b)$

$$f_X(x) = \begin{cases} \frac{1}{b-a} a \le x \le b \\ 0 \quad \text{otherwise} \end{cases}, \quad E(x) = \frac{a+b}{2}, \quad V(X) = \frac{(b-a)^2}{12}$$

$$F_X(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \le x \le b \\ 1 & x > b \end{cases}$$

Exponential Distribution, $X \sim \text{Exp}(\lambda)$

X be waiting time until the next event when events occur independently at a constant average rate. For $\lambda > 0$, λ is the average event rate, Mean waiting time is $E(X) = \frac{1}{\lambda}$.

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \ge 0 \\ 0 & \text{if } x < 0 \end{cases}, \quad E(X) = \frac{1}{\lambda}, \quad V(X) = \frac{1}{\lambda^2}$$

$$F_X(x) = P(X \le x) = \begin{cases} 1 - e^{-\lambda x} & \text{if } x \ge 0 \\ 0 & \text{if } x < 0 \end{cases}$$

Hence, for $x \ge 0$,

$$P(X > x) = e^{-\lambda x}$$

Let $\mu = \frac{1}{\lambda}$. Then

$$f_X(x) = \begin{cases} \frac{1}{\mu} e^{-\frac{x}{\mu}} & \text{if } x \ge 0 \\ 0 & \text{if } x < 0 \end{cases}$$

$$E(X) = \mu, \quad V(X) = \mu^2, \quad F_X(x) = 1 - e^{-\frac{x}{\mu}}$$

Normal Distribution, $X \sim N(\mu, \sigma^2)$

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, -\infty < x < \infty, E(X) = \mu, V(X) = \sigma^2$$

$$Z = \frac{X - \mu}{\sigma}$$

$$\Phi(z) = F_Z(z) = P(Z < z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

$$P(x_1 < X < x_2) = \Phi\left(\frac{x_2 - \mu}{\sigma}\right) - \Phi\left(\frac{x_1 - \mu}{\sigma}\right)$$

$$P(Z \leftarrow z) = P(Z \ge z) = 1 - \Phi(z)$$

Quartile α

$$P(Z \ge z_\alpha) = \alpha$$

Approximate Binomial using Normal Distribution

As $n \to \infty$ and p is constant, we may approximate binomial distribution using normal distribution.

$$Z = \frac{X - E(X)}{\sqrt{V(X)}} = \frac{X - np}{\sqrt{np(1-p)}}$$

Use when $np > 5$ and $n(1-p) > 5$

Continuity Correction

$$P(X = k) \approx P\left(k - \frac{1}{2} < X < k + \frac{1}{2}\right)$$

$$P(a \le X \le b) \approx P\left(a - \frac{1}{2} < X < b + \frac{1}{2}\right)$$

$$P(a < X \le b) \approx P\left(a + \frac{1}{2} < X < b + \frac{1}{2}\right)$$

$$P(a \le X < b) \approx P\left(a - \frac{1}{2} < X < b - \frac{1}{2}\right)$$

$$P(a < X < b) \approx P\left(a + \frac{1}{2} < X < b - \frac{1}{2}\right)$$

$$P(X \le c) = P(0 \le X \le c) \approx P\left(-\frac{1}{2} < X < c + \frac{1}{2}\right)$$

$$P(X > c) = P(c < X \le n) \approx P\left(c + \frac{1}{2} < X < n + \frac{1}{2}\right)$$